

5

NUMBER AND ALGEBRA

ALGEBRA AND EQUATIONS

Algebra is the branch of mathematics that uses letters of the alphabet to write general rules called **formulas** or **equations**. Algebra has been a powerful tool for mathematicians for over 4000 years and was first used in ancient Ba ylon and Egypt in 2000 BCe It was brought to Europe from India by the Arabs during the 9th century, and the word *algebra* comes from the Arabic *al-jabr,* a method of solving equations. Algebra today has uses in technology-related careers such as science, medicine and finance.



Chapter outline

	Working mathematically				
5.01 The laws of arithmetic	U	F		R	
5.02 The distributive law	U	F		R	C
5.03 Variables	U	F		R	C
5.04 Creating algebraic expressions	U	F		R	C
5.05 Substitution	U	F		R	
5.06 Collecting variables	U	F		R	C
5.07 Adding and subtracting terms	U	F		R	C
5.08 Multiplying terms	U	F		R	C
5.09 Dividing terms	U	F		R	C
5.10 Equations	U	F			
5.11 One-step equations	U	F		R	C
5.12 Two-step equations	U	F		R	C
5.13 Equation problems	U	F	PS	R	

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Wordbank

algebraic term A part of an algebraic expression, such as $3x$ in $4x^2 + 3x - 11$

evaluate To find the alue of an expression

formula A general mathematical rule written using letters and symbols

inverse operation An opposite operation used in solving an equation, for example, the inverse operation to multiplying is dividing

like terms Terms that have exactly the same variables, for example, $5p$ and $2p$

solve (an equation) To find the alue of an unknown variable in an equation

substitute To replace a variable with a number

variable A quantity that can take on different values, represented by a symbol such as a letter of the alphabet

Quiz
Wordbank 5

Videos (15):

5.02 Multiplying by 8, 9, 11 and 12
5.03 Algebraic expressions • Simplifying algebraic expressions 2
5.04 From words to algebraic expressions
5.05 Substitution 1 • Substitution 2
5.07 Adding and subtracting terms
5.08 Multiplying terms
5.09 Dividing terms
5.11 One-step equations
5.12 Two-step equations 1 • Two-step equations 2 • Solving equations
5.13 Equation problems 1 • Equation problems 2

Twig videos (2):

5.05 The heartbeat formula
5.09 The Arabic science of balancing

PhET interactives (5):

5.02 Area model: Introduction • Area model: Multiplication
5.07 Expression exchange
5.09 Equality explorer: Basics • Equality explorer

Quizzes (6):

- Wordbank 5
- SkillCheck 5
- Mental skills 5A
- Mental skills 5B
- Language of maths 5
- Test yourself 5

Skillsheets (3):

5.03 Order of operations
5.04 Algebraic expressions
5.07 Algebra using diagrams

Worksheets (13):

5.04 Writing algebraic expressions
5.05 Substitution • Why aren't they the same?
5.07 Collecting like terms
5.08 Perimeter and area
5.09 Simplifying algebraic expressions 1 • Why aren't they the same?
5.10 Guess-and-check
5.11 One-step equations
5.12 Equations writing activity • Equations match • Backtracking 1 • Backtracking 2

Mind map Algebra and equations

Puzzles (11):

5.03 Algebra bingo game
5.04 What's the expression? • Generalised arithmetic • What does the symbol mean?
5.05 Substitution game • Formula 1 game
5.07 Algebra bingo game • Adding and subtracting like terms
5.11 Equations dominoes
5.12 Solving equations
5.13 Writing and solving equations

Language of maths Algebra find-a-word

Presentation (1):

5.07 Collecting like terms

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- use the laws of arithmetic (commutative, associative, distributive) to calculate with numbers mentally and with pen-and-paper
- use algebra and variables to generalise the laws of arithmetic
- use variables to write algebraic terms and expressions
- convert between worded and algebraic expressions
- evaluate algebraic expressions by substituting values
- add, subtract, multiply and divide algebraic terms
- solve one- and two-step linear equations
- apply equations to solve real-life problems

SkillCheck ANSWERS ON P. 574



1 If ● = 3, then find the value of each expression.

a ● + 5	**b** 7 − ●	**c** 2 × ●	**d** 12 + ●
e ● − 1	**f** ● × 10	**g** ● + ●	**h** 27 ÷ ●
i −6 + ●	**j** ● × (−1)	**k** −3 ÷ ●	**l** −4 − ●

2 Find the value of ■ in each equation.

a ■ + 4 = 10	**b** 6 × ■ = 36	**c** ■ + ■ = 20
d 18 − ■ = 11	**e** 12 ÷ ■ = −3	**f** ■ − 6 = 1

3 Write the number that is:

a 4 more than 7	**b** double 10
c the next odd number after 9	**d** half of 18
e 10 less than 14	**f** the difference between 8 and 1
g 5 multiplied by itself	**h** 5 divided by itself
i the amount that 11 is greater than 9	**j** the product of 4 and 8

4 Evaluate each product mentally.

a 15 × 10	**b** 28 × 9	**c** 7 × 4 × 5
d 16 × 5	**e** 13 × 10	**f** 13 × 2

5 Is 13 × 12 is the same as 13 × 10 + 13 × 2?

6 Find the value of ▲ in each equation.

a ▲ − 7 = 5	**b** ▲ ÷ 5 = 4	**c** ▲ × ▲ = 81
d 33 ÷ ▲ = 3	**e** −4 + ▲ = 3	**f** ▲ − 8 = −3

7

a If ▲ = −7, evaluate 2 × ▲ + 3	**b** If ● = 4, evaluate 3 × ● + 5
c If ● = 15, evaluate ● ÷ 3 − 4	**d** If ▲ = 5, evaluate 4 × ▲ − 6
e If ■ = −2, evaluate 9 × 2 + ■	**f** If ■ = 6, evaluate 4 × 3 − ■
g If ● = −4, evaluate ● × 4 + 3	**h** If ▲ = −1, evaluate 10 × 2 − ▲

8 Evaluate each expression.

a 5 − 8	**b** −8 + 3	**c** 3 × (−4)	**d** −6 ÷ 3
e −2 + 5	**f** −2 − 9	**g** −4 × (−9)	**h** −12 ÷ (−4)
i −3 − 7	**j** 80 ÷ (−10)	**k** 6 − 15	**l** −8 × 9

5.01 The laws of arithmetic

Arithmetic is the branch of mathematics involved in calculating with **numbers**, involving the 4 operations of adding, subtracting, multiplying and dividing. We will now look at some general rules of arithmetic and use algebraic symbols to write these rules formally.

The commutative laws

The **commutative law** says that you can operate with 2 numbers in reverse order and still get the same answer. For example:

$4 + 7 = 11$ and $7 + 4 = 11$, so $4 + 7 = 7 + 4$.

$9 \times 5 = 45$ and $5 \times 9 = 45$, so $9 \times 5 = 5 \times 9$.

The commutative law works for **adding** or **multiplying** only. It does not work for subtracting or dividing. For example, $8 - 2$ is not equal to $2 - 8$ and $14 \div 2$ is not equal to $2 \div 14$.

The commutative laws of addition and multiplication

If a and b stand for numbers, then:

$$a + b = b + a$$
$$a \times b = b \times a$$

The associative laws

The **associative law** says that when operating with more than 2 numbers, you can group them in any way and still get the same answer. For example:

$$\begin{aligned} 6 + 7 + 4 &= (6 + 7) + 4 \\ &= 13 + 4 \\ &= 17 \end{aligned}$$

and

$$\begin{aligned} 6 + 7 + 4 &= 6 + (7 + 4) \\ &= 6 + 11 \\ &= 17 \end{aligned}$$

so $(6 + 7) + 4 = 6 + (7 + 4)$.

$$\begin{aligned} 5 \times 3 \times 2 \times 2 &= (5 \times 3) \times 2 \times 2 \\ &= 15 \times 2 \times 2 \\ &= 30 \times 2 \\ &= 60 \end{aligned}$$

and

$$\begin{aligned} 5 \times 3 \times 2 \times 2 &= 5 \times (3 \times 2) \times 2 \\ &= 5 \times 6 \times 2 \\ &= 30 \times 2 \\ &= 60 \end{aligned}$$

so $(5 \times 3) \times 2 \times 2 = 5 \times (3 \times 2) \times 2$.

The associative law also works for **adding** or **multiplying** only, not for subtracting or dividing. For example, $10 - 3 - 1$ is not equal to $10 - (3 - 1)$ and $20 \div 5 \div 2$ is not equal to $20 \div (5 \div 2)$.

The associative laws of addition and multiplication

If a, b and c stand for numbers, then:

$$(a + b) + c = a + (b + c)$$
$$(a \times b) \times c = a \times (b \times c)$$

The commutative and associative laws of arithmetic help us calculate mentally with numbers more easily.

Example 1

Evaluate each sum.

a $6 + 16$ **b** $26 + 15 + 4$ **c** $58 + 55 + 45$ **d** $-8 + 22$

Solution

a $6+16=16+6$
$=22$

It is easier to add smaller numbers to larger ones, so swap 6 and 16 around.

b When adding, pair numbers together that add to multiples of 10, 100, and so on.

$26+15+4=(26+4)+15$
$=30+15$
$=45$

26 and 4 add to 30, so put them together.

c $58+55+45=58+(55+45)$
$=58+100$
$=158$

55 and 45 add to 100, so put them together.

d $-8+22=22+(-8)$
$=22-8$
$=14$

It is easier to work with larger and positive numbers first, so swap –8 and 22 around.

Example 2

Evaluate each product.

a $17 \times 5 \times 2$ **b** $25 \times 9 \times 4$

Solution

a $17\times5\times2=17\times(5\times2)$
$=17\times10$
$=170$

$5 \times 2 = 10$, so put 5 and 2 together.

b $25\times9\times4=(25\times4)\times9$
$=100\times9$
$=900$

$25 \times 4 = 100$, so put 25 and 4 together.

Multiplying by a multiple of 10

Example 3

Evaluate each product.

a 9×50 **b** 6×400 **c** 20×300

Solution

You can split multiples of 10, 100, 1000 and so on to make multiplying easier.

a $9 \times 50 = 9 \times 5 \times 10$
$= 45 \times 10$
$= 450$

b $6 \times 400 = 6 \times 4 \times 100$
$= 24 \times 100$
$= 2400$

c $20 \times 300 = 2 \times 10 \times 3 \times 100$
$= 2 \times 3 \times 10 \times 100$
$= 6 \times 1000$
$= 6000$

EXERCISE 5.01 ANSWERS ON P. 574

The laws of arithmetic U F R

EXAMPLE 1

1 Evaluate each sum. R

a $4 + 21$ **b** $19 + 99$ **c** $18 + 23 + 7$
d $75 + 50 + 25$ **e** $16 + 80 + 44 + 10$ **f** $56 + 5 + 15 + 4$
g $-18 + 28$ **h** $16 + 45 + 4$ **i** $-6 + 20$
j $81 + 7 + 19 + 12$ **k** $54 + 27 + 16 + 3$ **l** $44 + 12 + 16 + 12$

EXAMPLE 2

2 Evaluate each product. R

a $2 \times 17 \times 5$ **b** $81 \times 2 \times 5$ **c** $50 \times 14 \times 2$
d $10 \times 32 \times 2$ **e** $5 \times 7 \times 4$ **f** $2 \times 5 \times 3 \times 5$
g $8 \times 3 \times 2$ **h** $25 \times 8 \times 2$ **i** $4 \times 4 \times 5$
j $20 \times 6 \times 5$ **k** $3 \times 2 \times 5 \times 3$ **l** $2 \times 14 \times 5 \times 10$

EXAMPLE 3

3 Evaluate each product. R

a 7×1000 **b** 100×100 **c** 3×90 **d** 4×700
e 5×80 **f** 2×600 **g** 4×40 **h** 12×2000
i 4×6000 **j** 3×1100 **k** 90×20 **l** 5×500
m 8×40 **n** 20×20 **o** 300×70 **p** 700×300

4 Evaluate $18 \times 4 \times 5$. Select the correct answer **A**, **B**, **C** or **D**. R

A 110 **B** 162 **C** 180 **D** 360

5 Evaluate 60×600. Select the correct answer **A**, **B**, **C** or **D**. R

A 1200 **B** 3600 **C** 12 000 **D** 36 000

6 Write True (T) or False (F) for each equation. R

a $4 - 8 = 8 - 4$ **b** $6 \times 5 = 5 \times 6$

c $10 + 11 = 11 + 10$ **d** $12 \div 4 = 4 \div 12$

e $a + 5 = 5 + a$ **f** $3 - x = x - 3$

g $b \div 2 = 2 \div b$ **h** $c \times 2 = 2 \times c$

i $x - y = y - x$ **j** $a \times b = b \times a$

k $x + y = y + x$ **l** $c \div d = d \div c$

The distributive law 5.02

The **distributive law** says that you can multiply by a number by splitting it into the sum or difference of 2 other numbers. Look at these examples.

Interactives
Area model: Introduction
Area model: Multiplication

a $17 \times 12 = 17 \times (10 + 2)$

$= (10 + 2) + (10 + 2) + (10 + 2) + \ldots$ 17 times

$= 10 + 10 + 10 + \ldots + 2 + 2 + 2 + \ldots$ 17 times each

$= (17 \times 10) + (17 \times 2)$

$= 170 + 34$

$= 204$

so $17 \times (10 + 2) = 17 \times 10 + 17 \times 2$

b $14 \times 9 = 14 \times (10 - 1)$

$= (10 - 1) + (10 - 1) + (10 - 1) + \ldots$ 14 times

$= 10 + 10 + 10 + \ldots - 1 - 1 - 1 - \ldots$ 14 times each

$= (14 \times 10) - (14 \times 1)$

$= 140 - 14$

$= 126$

so $14 \times (10 - 1) = 14 \times 10 - 14 \times 1$

The distributive law of arithmetic

If a, b and c stand for numbers, then:

$$a \times (b + c) = a \times b + a \times c$$

$$a \times (b - c) = a \times b - a \times c$$

The distributive law of arithmetic can help us multiply numbers more easily.

Foundation Standard Complex

Video
Multiplying by 8, 9, 11 and 12

Example 4

Evaluate each product.

a 25×9 **b** 18×8

Solution

a
$$\begin{aligned} 25 \times 9 &= 25 \times (10 - 1) \\ &= 25 \times 10 - 25 \times 1 \\ &= 250 - 25 \\ &= 225 \end{aligned}$$

b
$$\begin{aligned} 18 \times 8 &= 18 \times (10 - 2) \\ &= 18 \times 10 - 18 \times 2 \\ &= 180 - 36 \\ &= 144 \end{aligned}$$

Multiplying by 9 or 8

- To multiply a number by 9, multiply by 10, then subtract the number
- To multiply a number by 8, multiply by 10, then subtract double the number

Video
Multiplying by 8, 9, 11 and 12

Example 5

Evaluate each product.

a 13×11 **b** 27×12

Solution

a
$$\begin{aligned} 13 \times 11 &= 13 \times (10 + 1) \\ &= 13 \times 10 + 13 \times 1 \\ &= 130 + 13 \\ &= 143 \end{aligned}$$

b
$$\begin{aligned} 27 \times 12 &= 27 \times (10 + 2) \\ &= 27 \times 10 + 27 \times 2 \\ &= 270 + 54 \\ &= 324 \end{aligned}$$

Multiplying by 11 or 12

- To multiply a number by 11, multiply by 10 and then add the number
- To multiply a number by 12, multiply by 10 and then add double the number

EXERCISE 5.02 ANSWERS ON P. 575

The distributive law U F R C

1 Copy and complete each multiplication. R C

a
$$\begin{aligned} 18 \times 12 &= 18 \times (10 + ___) \\ &= 18 \times ___ + 18 \times ___ \\ &= _____ + 36 \\ &= _____ \end{aligned}$$

b
$$\begin{aligned} 16 \times 9 &= ___ \times (___ - 1) \\ &= ___ \times 10 - ___ \times 1 \\ &= 160 - ____ \\ &= _____ \end{aligned}$$

Foundation Standard Complex

9780170477963

c $21 \times 11 = 21 \times (___ + ___)$

$= ___ \times 10 + 21 \times ___$

$= 210 + ____$

$= _____$

d $15 \times 8 = 15 \times (10 - ___)$

$= ___ \times 10 - ___ \times ___$

$= ____ - ____$

$= _____$

2 Evaluate each product.

EXAMPLE 4

5.02

a 15×9	**b** 18×9	**c** 26×9	**d** 31×9
e 16×8	**f** 13×8	**g** 22×8	**h** 14×8
i 27×9	**j** 45×8	**k** 38×8	**l** 29×9

3 Dean tried to use the distributive law to evaluate 101×56. Find the error in his incorrect working. Select the correct answer **A**, **B**, **C** or **D**. R C

EXAMPLE 5

$101 \times 56 = 56 \times (100 + 1)$ **A**

$= 56 \times 100 + 1$ **B**

$= 5600 + 1$ **C**

$= 5601$ **D**

4 Evaluate each product.

a 17×11	**b** 24×11	**c** 15×11	**d** 29×11
e 25×12	**f** 14×12	**g** 24×12	**h** 33×12
i 40×12	**j** 12×11	**k** 9×12	**l** 19×11

5 Use the distributive law to copy and complete each multiplication. R C

a $6 \times 22 = 6 \times (20 + ___)$

$= 6 \times ___ + 6 \times ___$

$= 120 + _____$

$= _____$

b $12 \times 19 = ___ \times (20 - ___)$

$= ___ \times 20 - ___ \times 1$

$= 240 - ____$

$= _____$

c $40 \times 41 = ____ \times (___ + 1)$

$= ___ \times 40 + ____ \times ___$

$= 1600 + ____$

$= _____$

d $25 \times 48 = ____ \times (50 - ___)$

$= ___ \times 50 - ___ \times ___$

$= ____ - ____$

$= _____$

6 Use the distributive law to evaluate each product.

a 5×32	**b** 4×51	**c** 12×99	**d** 8×102
e 6×29	**f** 12×21	**g** 15×18	**h** 11×49

Foundation Standard Complex

7 Which is the correct distributive law for calculating 14×12?
Select the correct answer **A**, **B**, **C** or **D**.

A $14 \times 10 \times 2$ **B** $14 \times 10 + 2$

C $10 + 4 \times 10 + 2$ **D** $14 \times 10 + 14 \times 2$

5.03 Variables

Algebra uses letters or symbols to write general mathematical rules called **formulas** or **equations**. We have used algebra to write the laws of arithmetic we have learned.

The commutative laws:

$a + b = b + a$

$a \times b = b \times a$

The associative laws:

$(a + b) + c = a + (b + c)$

$(a \times b) \times c = a \times (b \times c)$

The distributive laws:

$a \times (b + c) = a \times b + a \times c$

$a \times (b - c) = a \times b - a \times c$

In the formulas above, a, b and c represent any 3 numbers and are called **variables** or **pronumerals**. A variable is a symbol, usually a letter of the alphabet, that represents a number. It is called a **variable** because its value can vary (change), and a **pronumeral** because it stands in place of a numeral.

Example 6

Examine each number pattern, then write the general rule using a variable.

a
$4 \times 1 = 4$
$7 \times 1 = 7$
$9 \times 1 = 9$
$1 \times 1 = 1$
$6 \times 1 = 6$
$b \times 1 =$ ____

b
$9 + 9 + 9 = 3 \times 9$
$4 + 4 + 4 = 3 \times 4$
$11 + 11 + 11 = 3 \times 11$
$2 + 2 + 2 = 3 \times 2$
$5 + 5 + 5 = 3 \times 5$
$x + x + x =$ ____

c
$6 - 6 = 0$
$1 - 1 = 0$
$17 - 17 = 0$
$12 - 12 = 0$
$4 - 4 = 0$

d
$2 \times (-2) = -(2^2)$
$8 \times (-8) = -(8^2)$
$5 \times (-5) = -(5^2)$
$9 \times (-9) = -(9^2)$
$1 \times (-1) = -(1^2)$

Foundation Standard Complex

9780170477963

Solution

The general rule is:

a $b \times 1 = b$ — Any number multiplied by 1 is itself

b $x + x + x = 3 \times x$ — The sum of the same number 3 times is 3 times that number

c $a - a = 0$ — Other variables are possible, for example, $k - k = 0$ — Any number minus itself is 0

d $r \times (-r) = -(r^2)$ — Other variables are possible, for example, $y \times (-y) = -(y^2)$ — Any number multiplied by its opposite is equal to the negative of that number squared

Algebraic abbreviations

Mathematicians prefer to write expressions as simply as they can. For example:

$5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 + 5 = 9 \times 5$

and $2 \times 2 \times 2 \times 2 \times 2 = 2^5$.

When writing **algebraic expressions**, we use the following abbreviations:

- $3 \times k = 3k$ — for multiplication, we leave out the '$\times$' symbol
- $m \div 4 = \frac{m}{4}$ — for division, we write in fraction form
- $r \times r = r^2$ — for powers
- $1x = x$ — for multiplying by 1
- $c \times a \times 6 = 6ac$ — write the number first, then the variables in alphabetical order

Example 7

Simplify each expression.

a $d \times 8$ **b** $m + m + m + m + m$ **c** $k \div 2$

d $u \times 9 \times w$ **e** $6x + 2x$ **f** $3 \times p \times 2 \times h$

g $z \times z \times z$ **h** $3n - 2n$ **i** $y \times y \times 4$

Solution

a $d \times 8 = 8d$ — Number first

b $m + m + m + m + m = 5m$

c $k \div 2 = \frac{k}{2}$

d $u \times 9 \times w = 9uw$ — Number first, then variables in alphabetical order

e $6x + 2x = 8x$ — $(x + x + x + x + x + x) + (x + x) = 8$ lots of x

f $3 \times p \times 2 \times h = 3 \times 2 \times p \times h$
$= 6hp$

5.03

Videos
Algebraic expressions
Simplifying algebraic expressions 2

Puzzle
Algebra bingo game

g $z \times z \times z = z^3$

h $3n - 2n = 1n$ $(n + n + n) - (n + n) =$ One n left

$= n$ Just n: no need to write the '1'

i $y \times y \times 4 = 4y^2$ The '2' (squared) belongs to the y only

Example 8

Skillsheet Order of operations

Use order of operations to simplify each expression.

a $2 \times r + 4$ **b** $15 - d \div 3$ **c** $7 \div (n \times n) + 1$

Solution

a $2 \times r + 4 = 2r + 4$ × first, then +

b $15 - d \div 3 = 15 - (d \div 3)$ ÷ first, then –

$= 15 - \frac{d}{3}$

c $7 \div (n \times n) + 1 = 7 \div n^2 + 1$ () first, then ÷, then +

$= \frac{7}{n^2} + 1$

Example 9

Write each expression in expanded form.

a $3ab$ **b** $-8st^2$ **c** $x^2 - \frac{y}{10}$

Solution

a $3ab = 3 \times a \times b$

b $-8st^2 = -8 \times s \times t \times t$

Note that expanding is the opposite of simplifying.

c $x^2 - \frac{y}{10} = x \times x - y \div 10$

EXERCISE 5.03 ANSWERS ON P. 575

Variables U F R C

EXAMPLE 6

1 Examine each number pattern, then write the general rule using a variable. A clue has been given in parts **a** and **b**. R C

a

$8 \div 8 = 1$

$1 \div 1 = 1$

$-2 \div (-2) = 1$

$14 \div 14 = 1$

$-7 \div (-7) = 1$

$u \div u =$ ____

b

$6 + 6 = 2 \times 6$

$10 + 10 = 2 \times 10$

$4 + 4 = 2 \times 4$

$9 + 9 = 2 \times 9$

$-4 + (-4) = 2 \times (-4)$

$p + p =$ ____

Foundation Standard Complex

9780170477963

c
$3 + (-3) = 0$
$7 + (-7) = 0$
$20 + (-20) = 0$
$1 + (-1) = 0$
$16 + (-16) = 0$

d
$2 \times 0 = 0$
$19 \times 0 = 0$
$-3 \times 0 = 0$
$0 \times 0 = 0$
$-24 \times 0 = 0$

e
$7 + 0 = 7$
$-7 + 0 = -7$
$3 + 0 = 3$
$11 + 0 = 11$
$-100 + 0 = -100$

f
$-5 \times (-5) = 5^2$
$-7 \times (-7) = 7^2$
$-12 \times (-12) = 12^2$
$-1 \times (-1) = 1^2$
$-8 \times (-8) = 8^2$

2 Simplify each expression.

EXAMPLE 7

a $w \times 3$
b $b \times b$
c $9 \times c \times a \times d$
d $a + a + a + a + a + a$
e $e \div 5$
f $f + f$
g $5m + 2m$
h $2 \times w \times 3 \times h$
i $5m - 2m$
j $4 \times f \times f$
k $6d - 5d$
l $a \times c \times b \times 10$
m $16 \div n$
n $g \times 2 \times g$
o $3x + 6x$
p $h + h - h$
q $12q - 4q$
r $2 \times m \times 4 \times n$
s $6 \times p \times r \times (-2) \times r$
t $a + a + a + b + b$
u $a \times a \times a \times b \times b$

3 Which of the following is not equal to $2p$? Select the correct answer **A**, **B**, **C** or **D**. R

EXAMPLE 8

A $3p - p$ **B** p^2 **C** $p + p$ **D** $p \times 2$

4 Use order of operations to simplify each expression. R

a $16 \div (a \times 5)$
b $7 + 3 \times n$
c $(e - 6) \div 2$
d $4 \times s - 8$
e $c \times c + d \times d$
f $k \div 9 + m$
g $k \div (9 + m)$
h $12 - r \div 2$
i $7 + u \times u \times 7$

5 Which of the following is equal to $m \times m \times 3 + 4$? Select the correct answer **A**, **B**, **C** or **D**.

A $3m^2 + 4$ **B** $2m^3 + 4$ **C** $6m + 4$ **D** $m^2 + 7$

6 Explain the meaning of: C

a xy **b** $\frac{2n}{r}$ **c** $1x = x$

EXAMPLE 9

7 Write each expression in expanded form. C

a $12rs$ **b** $-2n^2$ **c** $\frac{8}{f}$ **d** $r^2 + t^2$

e $9a^2b$ **f** $5mn - 2a$ **g** $\frac{x+1}{3}$ **h** q^3

i $14 - 2d^2$ **j** $\frac{-4c}{5}$ **k** x^2y^2 **l** $4j - \frac{8}{k}$

8 What is $4m^3$ in expanded form? Select the correct answer **A**, **B**, **C** or **D**.

A $4m \times 4m \times 4m$ **B** $4m + 4m + 4m$

C $4 \times m + m + m$ **D** $4 \times m \times m \times m$

Investigation

The laws of arithmetic

We can use algebra and variables to describe general laws about arithmetic.

For example, if we add zero to any number, the answer is still that number.

If we let N stand for any number: $N + 0 = N$

In groups of 2 to 4 students, answer the following questions.

1 Describe in words what each number rule below means.

a $N \times 1 = N$ **b** $N \times 0 = 0$

c $a - b \neq b - a$ **d** $a + b + c = b + a + c$

e $N - 0 = N$ **f** $ab = ba$

2 Write each rule algebraically using variables.

a Any number divided by 1 equals itself.

b Multiplying a number by 8 is the same as doubling it 3 times.

c Any 3 numbers can be multiplied together in any order.

d Any number added to itself is the same as multiplying that number by 2.

e Any number subtracted from itself equals 0.

f Any number multiplied by its reciprocal equals 1.

3 Is each equation true or false? Test your decision by substituting a number for the variable and checking.

a $a \div b = b \div a$ **b** $N \div N = 1$

c $4a - a = 4$ **d** a is a factor of a

e If N is even, then $N + 3$ is odd **f** $\frac{1}{2}N = N - 2$

g $a + (-a) = 2a$ **h** $0 \div N = 0$

i $N \div 0 = 0$ **j** $N \div 1 = N$

4 If k is an odd number, what is an expression for:

a the previous odd number? **b** the next even number?

Foundation Standard Complex

9780170477963

Creating algebraic expressions 5.04

We can use variables to convert worded expressions to algebraic expressions.

Example 10

If A represents any number, write an algebraic expression for:

a 3 times that number

b 3 less than that number

c the next consecutive number

d that number multiplied by itself

e the square root of that number

f one-third of that number

consecutive means 'following in order': for example, 7, 8, 9 are consecutive numbers

Solution

a $3 \times A = 3A$

b $A - 3$

c $A + 1$

The number after A is $A + 1$

(the number after 4 is $4 + 1 = 5$)

d $A \times A = A^2$

e $\sqrt{A}$

f $\frac{1}{3}A = A \div 3 = \frac{A}{3}$

Example 11

Write an expression for:

a the sum of m and 5

b double k

c u increased by 10

d the difference between p and q

e the product of r and 12

f how many times y divides into 50

Solution

a $m + 5$

b $2k$

c $u + 10$

d $p - q$

e $12r$

f $50 \div y = \frac{50}{y}$

Video
From words to algebraic expressions

Skillsheet
Algebraic expressions

Worksheet
Writing algebraic expressions

Puzzles
What's the expression?

Generalised arithmetic

What does the symbol mean?

5.04

EXERCISE 5.04 ANSWERS ON P. 575

Creating algebraic expressions U F R C

EXAMPLE 10

1 If N represents any number, write an algebraic expression for: R C

a the number added to itself
b half the number
c triple the number
d 7 less than the number
e one-tenth of the number
f the previous consecutive number
g 5 times the number
h the sum of the number and 21
i the difference between the number and 1
j the number divided by 8
k the number increased by 3
l the number cubed
m the square root of the number

2 Suppose that in question **1**, the variable X represents any number instead of N. What difference would this make to your answers? Does it matter which letter of the alphabet you choose to use? R C

EXAMPLE 11

3 Write an expression for: R C

a the sum of all x, y and z
b the difference between b and c
c the product of u and v
d the product of u, v and v
e the quotient of C and d
f the sum of m and n, divided by 2
g the square root of the product of 10 and a
h 3 less than b
i 5 decreased by c
j c decreased by 5
k how many times 3 divides into e
l t squared
m the product of x, z and 9
n how much k is more than 12
o half of the difference between 20 and g

4 What is an algebraic expression for 'the product of 4 and 3 more than a number? Select the correct answer **A**, **B**, **C** or **D**. R C

A $4n + 3$ **B** $4 \times 3 + n$ **C** $4 \times (n + 3)$ **D** $4 \times 3 \times n$

5 Write an expression for: R C

a the number of students in a class if there are b boys and g girls
b the number of chicken nuggets needed for a party if there are n guests and each guest eats an average of 4 nuggets
c the number of children left in a class of d students if r of them have gone to the zone swimming carnival
d the amount of dollars earned selling s sushi rolls at the canteen, where each roll is sold at $2
e the cost of one film ticket if it costs $\$a$ for 3 tickets
f the perimeter of a rectangle of length x and width y
g the area of a rectangle of length x and width y
h the total cost of buying m milkshakes and i ice creams, where each milkshake costs $8 and each ice cream costs $5

Foundation Standard Complex

9780170477963

6 What is the correct expression for the total admission cost to Water World for c children at \$55 each and a adults at \$77 each? Select the correct answer **A**, **B**, **C** or **D**. R C

A $55a + 77c$ **B** $132ac$ **C** $77a + 55c$ **D** $a + c + 132$

7 Write an expression for the length of one side of a square that has a perimeter of $3a + 2$ units.

Mental skills 5A: Maths without calculators ANSWERS ON P. 576

5.04
Quiz
Mental skills 5A

Multiplying an even number by 5

To multiply an even number by 5, it's easier to halve it, then multiply by 10 (by inserting a 0 at the end). This is because $\frac{1}{2} \times 10 = 5$.

1 Study each example.

a $14 \times 5 = 14 \times \frac{1}{2} \times 10 = 7 \times 10 = 70$

b $36 \times 5 = 36 \times \frac{1}{2} \times 10 = 18 \times 10 = 180$

c $22 \times 5 = 11 \times 10 = 110$

d $18 \times 5 = 9 \times 10 = 90$

2 Now simplify each product.

a 32×5 **b** 26×5 **c** 12×5 **d** 28×5

e 42×5 **f** 54×5 **g** 38×5 **h** 44×5

i 60×5 **j** 34×5 **k** 16×5 **l** 58×5

Technology

Using variables and values

1 a Create the following spreadsheet to calculate the value of the expression $x + 2$ for different values of x.

	A	B	C	D	E	F	G	H	I	J	K
1	x	1	2	3	4	5	10	20	25	50	100
2	x + 2										
3											

b To evaluate $x + 2$ when $x = 1$ (column B), enter **=B1+2** into cell B2. What value should you get?

	A	B	C	D	E	F	G	H	I	J	K
1	x	1	2	3	4	5	10	20	25	50	100
2	x + 2	=B1+2									
3											

c To complete the table, use **Fill Right** to copy the rule from cell B2 across row 2 until cell K2. **Centre** your answers in each of the cells. Check that the answers look correct.

d Type over some values in row A of the spreadsheet to find the value of $x + 2$ when:

i $x = 6$ **ii** $x = 17$ **iii** $x = 31$

e Predict what value of x will give the answer 96 and check your guess using the spreadsheet.

Foundation Standard Complex

2 **a** Create the following spreadsheet to calculate the value of $x - 3$ for values of x.

	A	B	C	D	E	F	G	H	I	J	K
1	x	1	2	3	4	5	10	20	25	50	100
2	x - 3										

b In cell B2, enter a formula beginning with **=B1** to calculate $x - 3$.

c Use **Fill Right** to copy this formula up to cell L2. **Centre** your answers.

d Type over some values in row A to evaluate $x - 3$ when:

i $x = 9$ **ii** $x = 28$

e Predict what value of x will give the answer:

i 37 **ii** 105

Check your guess using the spreadsheet.

3 **a** Create the following spreadsheet and in cell B2, enter a formula to evaluate $x \div 2$.

	A	B	C	D	E	F	G	H	I	J	K
1	x	-8	-6	-4	-2	0	2	10	20	30	40
2	x ÷ 2										

b Use **Fill Right** to copy this formula up to cell L2. **Centre** your answers.

c Use the spreadsheet to evaluate $x \div 2$ when:

i $x = -10$ **ii** $x = 6$

d Predict what value of x will give the answer 21 and check using the spreadsheet.

4 **a** Create the following spreadsheet and in cell B2, enter a formula to evaluate $10x - 1$.

	A	B	C	D	E	F	G
1	x	-10	-5	0	5	10	15
2	10x - 1						

b Use **Fill Right** to copy this formula up to cell L2. **Centre** your answers.

c Use the spreadsheet to evaluate $x \div 2$ when:

i $x = 8$ **ii** $x = -7$

d Predict what value of x will give the answer 199 and check using the spreadsheet.

5 **a** Create a new spreadsheet to evaluate x^2 for different values of x. In cell A1, enter the label 'x', then along row A enter the following values: –2, –1, 0, and so on until 4 in cell H1.

b In cell A2, enter the label 'x^2' and in cell B2, enter a formula to evaluate x^2.

c **Fill Right** to complete the answers for cells C2 to H2.

Substitution

5.05

The word **substitution** means replacing one thing with another thing. In sport, substitution means to replace one player with another during a game.

In algebra, substitution means to replace a variable with a value to evaluate an algebraic expression.

Worksheet
Substitution

Puzzles
Substitution game
Formula 1 game

Videos
Substitution 1
Substitution 2
The heartbeat formula

Example 12

a Evaluate $4k - 9$ when $k = 15$

b If $r = m^2 + 3m$, evaluate r when $m = 4$

c If $a = 6$, $b = -4$ and $c = 10$, evaluate $\frac{c-a}{b}$.

Solution

a When $k = 15$,

$4k - 9 = 4 \times 15 - 9$

$= 51$

b When $m = 4$,

$r = 4^2 + 3 \times 4$

$= 28$

c When $a = 6$, $b = -4$ and $c = 10$,

$\frac{c-a}{b} = \frac{10-6}{-4}$

$= \frac{4}{-4}$

$= -1$

Substituting into formulas

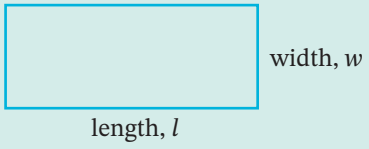
A **formula** is a general rule written as an algebraic equation showing the relationship between variables. Solving mathematical problems often involves substituting values into formulas.

Example 13

The formula for the perimeter of a rectangle is

$$P = 2l + 2w$$

where P is the perimeter, l is the length and w is the width.



Use this formula to calculate the perimeter of a rectangle with length 16 m and width 3 m.

Solution

When length $l = 16$ and width $w = 3$,

$P = 2l + 2w$

$= 2 \times 16 + 2 \times 3$

$= 38$ m

EXERCISE 5.05 ANSWERS ON P. 576

Substitution U F R

EXAMPLE 12

1 Evaluate $k + 3$ when:

a $k = 12$ **b** $k = 48$ **c** $k = 119$ **d** $k = -21$

2 Evaluate $45 - k$ when:

a $k = 5$ **b** $k = -13$ **c** $k = 28$ **d** $k = 45$

3 Evaluate $4k$ when:

a $k = 2$ **b** $k = 11$ **c** $k = -4$ **d** $k = \frac{1}{2}$

4 Find the value of $5k + 1$ when:

a $k = 3$ **b** $k = -4$ **c** $k = 20$ **d** $k = 7$

5 Find the value of $9k - 8$ if:

a $k = 9$ **b** $k = -5$ **c** $k = 0$ **d** $k = 2$

6 Find the value of $\frac{k}{3}$ if:

a $k = 15$ **b** $k = 33$ **c** $k = -9$ **d** $k = 57$

7 If $a = 6$, $b = 14$ and $c = -2$, find:

a $a + b - c$ **b** $2b + c$ **c** $\frac{a-b}{c}$ **d** $ab - 50$

8 If $m = 4$, $n = -2$ and $p = 3$, find:

a m^2 **b** $3n + p$ **c** mnp

9 If $d = 4c - 10$, find d when:

a $c = 5$ **b** $c = 8$ **c** $c = -10$

10 If $b = 3t - 1$, find b when:

a $t = 5$ **b** $t = 12$ **c** $t = -9$

11 If $y = 4 - ax^2$, find the value of y when $a = -2$ and $x = -3$.
Select the correct answer **A**, **B**, **C** or **D**.

A -54 **B** 54 **C** 22 **D** -14

12 Test whether $4x(x - 3) = 4x^2 - 12x$ by substituting $x = 4$ and $x = 11$. Is the expression true or false? R

13 Complete the table below for the expression $2x - 1$.

x	0	1	2	3	4	10
$2x - 1$						

EXAMPLE 13

14 The formula for the cost of a party is

$$C = 45n + 500$$

where C is the cost in dollars and n is the number of guests.

Use this formula to calculate the cost of a party for 72 guests.

Foundation Standard Complex

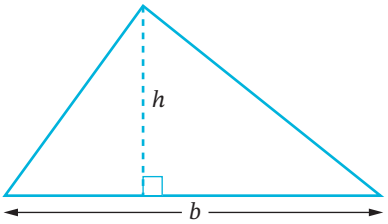
9780170477963

15 The formula for the area of a triangle is

$$A = \frac{1}{2}bh$$

where A is the area of the triangle, b is the length of the base and h is the height.

Find the area of a triangle with base length 9 m and height 6 m.



16 The number of hours sleep recommended for children has the formula

$$H = 17 - \frac{A}{2}$$

where H is the number of hours and A is the age of the child.

Find the number of hours sleep recommended for a 13-year-old person.



Shutterstock.com/FamVeld

17 The formula for converting Celsius temperatures to Fahrenheit temperatures is

$$F = \frac{9}{5}C + 32$$

where F is the temperature in Fahrenheit and C is the temperature in Celsius.

Use the formula to convert 30°C to Fahrenheit.

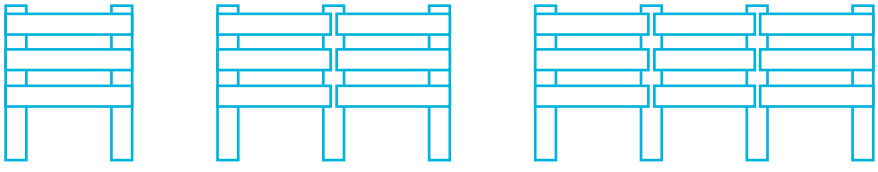
18 A locksmith charges according to this formula

$$C = 95h + 100$$

where C is the charge in dollars and h is the number of hours worked.

How much does the locksmith charge for working for 2 hours?

19 Krystal is building a timber fence at her farm. The sections of fence are shown below:



1 section, 5 pieces 2 sections, 9 pieces 3 sections, 13 pieces

The formula to calculate the number of pieces (P) to be used for a certain number of sections (S) is $P = 4S + 1$. How many pieces of timber are needed to make 16 sections? Select the correct answer **A**, **B**, **C** or **D**.

A 65 **B** 64 **C** 20 **D** 19

5.05

Foundation | Standard | Complex

Technology

Substitution

1 Create the spreadsheet below for evaluating 5 algebraic expressions involving two variables, a and b, whose values are entered in columns A and B.

	A	B	C	D	E	F	G
1	a	b	ab	a + b	2a + b	5(a + b)	3b ÷ 2a
2	2	1	=A2*B2				
3	10	0					
4	6	8					
5	-3	5					
6	0.4	1.7					
7	86	45					

2 Column C is for evaluating ab for the different values of a and b shown in rows 2 to 7. In cell C2, enter the formula for calculating ab using the values in A2 and B2.

3 **Fill Down** to copy this formula down to C7.

4 Enter appropriate formulas for the other 4 expressions in cells D2 to G2. Use brackets where necessary.

5 For each column, D to G, **Fill Down** to row 7.

6 Create the new spreadsheet below that shows a value for each variable, for example, $m = 5$, $n = 2$, $p = -4$.

	A	B	C	D	E	F	G
1	m	n	p	t	u	A	B
2	5	2	-4	12	0	50	31
3							

7 In each cell given, enter an appropriate formula to evaluate the algebraic expression shown.

a In cell A4: $m + p$

b In cell B4: npt

c In cell C4: $n + 2p$

d In cell D4: $B - A$

e In cell A5: $(p - n) \div t$

f In cell B5: $np + tu$

g In cell C5: $\frac{mt}{p}$

h In cell D5: $u - At$

i In cell A6: $A^2 - B^2$

j In cell B6: $6p + t$

k In cell C6: $\frac{m(t-n)}{A}$

l In cell D6: $3n^2 - p$

Investigation

Equivalent expressions

1 Determine whether each equation below is true or false by substituting a number (of your own choice) for the variable and checking whether the left expression equals the right expression.

a	$6 + 4m + 4 = 10 + 4m$	**b**	$5x + 3x = 8x^2$	**c**	$4k - k = 4$
d	$7g + 12 = 19g$	**e**	$3 + 7n + 2n = 3 + 9n$	**f**	$3p \times 7 = 21p$
g	$21m \div 3 = 7m$	**h**	$2y \times 4y = 24y$	**i**	$2r \times 2 \times 2r = 8r^2$

2 For each of the false statements in question **1**, work out what the correct statement should be.

Worksheet
Why aren't they the same?

Collecting variables 5.06

Let the variable x stand for an unknown number. Let x be represented by a cup that holds the unknown number of balls.

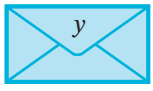


Then the total number of balls represented in the diagram below can be written using the algebraic expression $3 \times x + 5 = 3x + 5$.

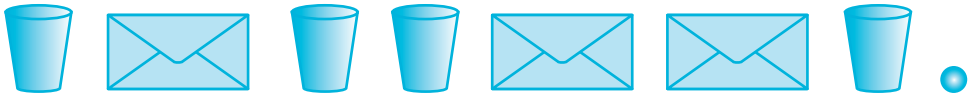


Note that even though we don't know how many balls are in each cup, it is the same number of balls, x, and we can still describe the total number of balls algebraically.

Let y stand for an unknown number that is different from x. Let y be represented by an envelope that contains this number of balls.



Then the total number of balls represented in the diagram below can be written using the algebraic expression $4x + 3y + 1$.



EXERCISE 5.06 ANSWERS ON P. 576

Collecting variables U F R C

1 Write an algebraic expression for the number of balls represented in each diagram, if each cup holds x balls. R C

a



b



c



d



e



f



g



h



i



j



2 Write an algebraic expression for the number of balls represented in each diagram, if each cup holds x balls and each envelope holds y balls. R C

a



b



c



d



e



f



g



h



3 Draw cups and balls to represent each algebraic expression, if each cup holds x balls. C

a $x + 3$

b $x + x + 1$

c $4 + 2x$

d $2 + x + 2$

e $x + 5 + 2x$

f $1 + 2x + 3 + x$

4 If N stands for the number of paperclips in each envelope, which diagram represents $3N + 2$ paperclips? Select the correct answer **A**, **B**, **C** or **D**. R C

A



B



C



D



5 If m stands for the number of coins in each envelope and p stands for the number of coins in each cup, match each expression in the left-hand column with the correct expression in the right-hand column. R C

a



$m + p + m$

b



$p + m + m + p$

c



$p + m + p + m + p$

d



$p + 2 + p + 1$

e



$2m + 2 + m + 1$

f



$2 + p + 2 + 2p$

A



$2m + 2p$

B



$2p + 3$

C



$3m + 3$

D



$p + 2m$

E



$3p + 4$

F



$3p + 2m$

6 Match each expression in the left column with the correct expression on the right. R C

a



$m + p - m$

b



$m + 4 - 2$

c



$2m + p - p$

A



$2m$

B



$2p + 2$

C



$2p$

Foundation | Standard | Complex

d



$2p + 6 - 4$

D



p

e



$2m + 2p - 2m$

E



$2p + 2m$

f



$2p + 3m - m$

F



$m + 2$

7 Simplify each expression. R C

a



$p + m + p$

b



$m + 2p - p$

c



$2p + 2 + p + 3$

d



$p + m + 2 + p + m - 1$

e



$3p + 2m - m - p$

8 Simplify $9a + 4b - 5a + 2b$. Select **A**, **B**, **C** or **D**. R

A $14a + 6b$ **B** $10ab$ **C** $4a + 6b$ **D** $14a + 11b$

9 Simplify each expression. R

a $3d + 2d + 4a$ **b** $6h + 4r + 5r - h$ **c** $e + e - e$

d $2k + 3 + 2k - 6$ **e** $3x + 4z - x + 2z$ **f** $2s - p - s + 3p$

g $6u + 4 - 2u + 8$ **h** $7n - 6 + 5 - 3n$ **i** $20 + 6i - 3i - 20$

Adding and subtracting terms

5.07

What is the fastest way to mentally calculate the sum $3 + 3 + 5 + 3 + 5 + 5 + 3 + 3 + 5$?

Do you think of collecting all the 3s together and all the 5s together like this?

$$\begin{aligned} 3 + 3 + 5 + 3 + 5 + 5 + 3 + 3 + 5 &= (3 + 3 + 3 + 3 + 3) + (5 + 5 + 5 + 5) \\ &= (5 \times 3) + (4 \times 5) \\ &= 15 + 20 \\ &= 35 \end{aligned}$$

We can do the same thing algebraically. Suppose an unknown number p is represented by a cup  and another unknown number m is represented by an envelope  . Now consider these 2 sums:

a



$p + m + p + m = 2p + 2m$

b



$m + p + m + p + m = 3m + 2p$

Note that we can add all the p **terms** together, and we can add all the m terms together, but we cannot add the ps and ms together because p and m represent different numbers. We cannot simplify the answers $2p + 2m$ and $3m + 2p$ because p and m are different.

These are examples of collecting **like terms**, terms that contain exactly the same variable(s), such as $4h$, $5h$ and h. Here are some more examples:

$3x$ and $-5x$	$4mw$ and $2mw$	$12m$ and m
$-6d$ and $10d$	$5ab$ and $2ba$	xyz and $2yzx$

Note: $ab = ba$

Here are some examples of **unlike terms**, terms that have different variables:

$3x$ and $5m$ $\quad$ $4mw$ and $2mn$ $\quad$ $-6d$ and $10d^2$

We can add or subtract like terms because the variables in them represent the same number. We can only add or subtract things that are the same. For example:



$3m + 2p - 2m = m + 2p$

$$\begin{aligned} (3 \text{ lots of } m) + (2 \text{ lots of } p) - (2 \text{ lots of } m) &= (3 \text{ lots of } m - 2 \text{ lots of } m) + (2 \text{ lots of } p) \\ &= (1 \text{ lot of } m) + (2 \text{ lots of } p) \end{aligned}$$

Skillsheet
Algebra using diagrams

Worksheet
Collecting like terms

Puzzles
Algebra bingo game

Adding and subtracting like terms

Interactive
Expression exchange

This makes sense because if $m = 5$ and $p = 8$:

$3m + 2p - 2m = (3 \times 5) + (2 \times 8) - (2 \times 5)$

$= (3 \times 5) - (2 \times 5) + (2 \times 8)$ grouping like terms

$= (1 \times 5) + (2 \times 8)$ 3 lots of 5 – 2 lots of 5 = 1 lot of 5

Adding and subtracting terms

- When adding and subtracting terms, only like terms can be collected
- Like terms have exactly the same variables

Example 14

Select the like terms from each list below.

a $5x, -x, x^2, xy, \frac{1}{2}x, -8x, \frac{4x}{3}$

b $3ab, 5, a, ba, -2b, 4ab^2, -ab$

Solution

a $5x, -x, xy, \frac{1}{2}x, -8x, \frac{4x}{3}$ are like terms because they each have a single x in them.

b $3ab$, ba and $-ab$ are like terms because each contains ab.

Example 15

Video Adding and subtracting terms

Presentation Collecting like terms

Simplify each expression by collecting like terms.

a $7d + 3e + 3d + e$

b $6m - 8 + 3m + 6$

c $6xy + 7x^2 - 3yx - 12x^2$

d $14 - 4t - 9t + 2$

Solution

a $7d + 3e + 3d + e = 7d + 3d + 3e + e$ Grouping like terms

$= 10d + 4e$

b $6m - 8 + 3m + 6 = 6m + 3m - 8 + 6$ The + or – sign goes with the term after it

$= 9m - 2$

c $6xy + 7x^2 - 3yx - 12x^2 = 6xy - 3xy + 7x^2 - 12x^2$ $3yx = 3xy$

$= 3xy - 5x^2$

d $14 - 4t - 9t + 2 = 14 + 2 - 4t - 9t$

$= 16 - 13t$

EXERCISE 5.07 ANSWERS ON P. 576

Adding and subtracting terms UFRC

EXAMPLE 14

1 Select the like terms from each list. C

a $4p, 2y, 2p, 5z, p$
b $m^2, m, n, 3m^2, mn$
c $5ac, 4a, 7ca, 3, 5$
d $vw, 5v, 9v, w, v^2, 2wv$
e $p, 6pq, 2qp, 3pq, 7q$
f $2d, 5, 3d^3, 7d^2, 9, 3d, 2$
g $4mn, 3m, 4, 2nm, mn, 2n$
h $x^2y, 2x, 3y, 4x^2y$
i $7, 2a, 4b, 5ba, 9, ab, 2$

2 What are the like terms in $3ab, 2a^2, 2b^2a, 2p^2, 2ba$?
Select the correct answer **A**, **B**, **C** or **D**. C

A $2a^2, 2p^2$
B $2a^2, 2b^2a, 2p^2, 2ba$
C $3ab, 2ba$
D $2b^2a, 2ba$

3 Simplify each expression by collecting like terms.

a $4k + 7k$
b $5mn + 2nm$
c $xy + xy$
d $3abc + 4abc + 2bac$
e $3x^2 + 2x^2 + x^2$
f $ef^2 + 4ef^2$
g $8d - 3d$
h $12mk - 7mk$
i $12xy - 7yx - xy$
j $8de - 12de$
k $6k^2 - 3k^2$
l $4w^2 - w^2$

EXAMPLE 15

4 Which expression is $2m + 4p - 5m - 2p$ simplified? Select **A**, **B**, **C** or **D**.

A $-mp$
B $7m + 6p$
C $-3m + 2p$
D $-3p$

5 Simplify each expression.

a $5k - 2j + 6j$
b $10ab + 2 - ab$
c $12m^2 - 5m^2 + 2m$
d $12s^2 - 7st - 2s^2$
e $4mn + 7mn - mn$
f $14k + 3 - 5k$
g $y^2 + 2y^2 - y^2$
h $9d - 4e + 3d$
i $7 - 2x - 3$
j $15gb - 8gb + gb$
k $4 - 2q + 5$
l $5a + 6c - 4c$

6 Simplify each expression.

a $4x - y - x - 2y$
b $7ab + 2bc - 3ab + bc$
c $4k^2 + 3 + 5k^2 + 8$
d $4bc - 8a + 2bc - a$
e $x^2 - 6x - x + 7$
f $p^2 + q^2 + 3p^2 - q^2$
g $3 + 4y + 8y + 11$
h $11r - 3s - 4r + 6s$
i $7g + 8b - b - g$
j $6x^2 - 4x - 9x + 6$
k $20a + 11m + 32a - 11m$
l $3q - 7 - 8q + 12q$
m $5 - 2y - 5 - 4y$
n $s^2 + 4s^2 + 12 - 5s^2$

Foundation | Standard | Complex

9780170477963

7 Which expression is $a^5 + a^5$ simplified? Select **A**, **B**, **C** or **D**.

A a^{10} **B** a^{25} **C** $2a^5$ **D** $2a^{10}$

8 Write an algebraic expression involving a sum that simplifies to: R C

a $2x + 6$ **b** $6m + 12p$ **c** $x^2 - x$

9 Write an algebraic expression involving a difference that simplifies to: R C

a $2x + 6$ **b** $6m + 12p$ **c** $x^2 - x$

5.08 Multiplying terms

Worksheet
Perimeter and area

The associative law of multiplication tells us that to calculate the product $6 \times 4 \times 5 \times 3$ mentally, we can rearrange the numbers to pair convenient factors like this:

$$\begin{aligned} 6\times4\times5\times3 &= (4\times5)\times(6\times3) \\ &= 20\times18 \\ &= 360 \end{aligned}$$

We can do the same thing algebraically. As numbers can be multiplied in any order, we can multiply like *and* unlike terms.

Video
Multiplying terms

Example 16

Simplify each product.

a $7b \times 3a$ **b** $8x \times (-2y)$ **c** $5d \times 6d \times 4$

d $9m^2 \times 4p$ **e** $4a \times 3b \times (-2a)$

Solution

a $7b \times 3a = 7 \times b \times 3 \times a$
$= 7 \times 3 \times a \times b$
$= 21ab$

Group numbers and variables separately.

b $8x \times (-2y) = 8 \times (-2) \times x \times y$
$= -16xy$

c $5d \times 6d \times 4 = 5 \times 6 \times 4 \times d \times d$
$= 120d^2$

d $9m^2 \times 4p = 9 \times 4 \times m^2 \times p$
$= 36m^2p$

e $4a \times 3b \times (-2a) = 4 \times 3 \times (-2) \times a \times a \times b$
$= -24a^2b$

Multiplying terms

When **multiplying algebraic terms**, multiply the numbers first, then multiply the variables. Write all variables in alphabetical order.

Foundation Standard Complex

9780170477963

Example 17

Use index notation to simplify each expression.

a $k \times k \times k \times k$ **b** $j \times j \times j \times j \times j \times j$

Solution

a $k \times k \times k \times k = k^4$ **b** $j \times j \times j \times j \times j \times j = j^6$

Video
Multiplying terms

5.08

EXERCISE 5.08 ANSWERS ON P. 576

Multiplying terms U F R C

EXAMPLE 16

1 Simplify $2p \times 5pq$. Select the correct answer **A**, **B**, **C** or **D**.

A $10pq$ **B** $7p^2q$ **C** $7pq$ **D** $10p^2q$

2 Simplify each expression.

a	$5 \times 3y$	**b**	$8 \times 2m$	**c**	$4k \times m$
d	$4t \times 5$	**e**	$2 \times f \times 6$	**f**	$2x \times 8y$
g	$10p \times 3m$	**h**	$3b \times 6d$	**i**	$3d \times 4d$
j	$5 \times 4m \times 6n$	**k**	$2y \times 3y \times 8$	**l**	$3r \times 3 \times 3r$
m	$2 \times 4a \times 9b$	**n**	$x \times xy \times y$	**o**	$2j \times 3k \times 7$
p	$5rst \times 2rs$	**q**	$3ac \times 6cd$	**r**	$hjk \times hj \times jk$
s	$4q^2 \times 6p^2$	**t**	$2m \times 3n \times mn$		

3 Simplify each expression.

a	$-4b \times 2d$	**b**	$6m \times (-4m)$	**c**	$-2 \times 5k \times 9l$
d	$20 \times (-3p)$	**e**	$9k \times (-7)$	**f**	$-15b \times (-5b)$
g	$-3m \times mn \times 4n$	**h**	$6abc \times (-2ab)$	**i**	$5r \times (-2r) \times (-4)$
j	$9d \times (-4e) \times 6d$	**k**	$-8x \times 9x \times (-2)$	**l**	$5m \times 4 \times (-2m)$

4 Which expression simplifies to $14a^2n^2$? Select **A**, **B**, **C** or **D**.

A $2a \times an \times 7$ **B** $2a \times 7n \times an$ **C** $7a^2 \times 2n$ **D** $7a^2n^2 \times 2a^2n^2$

EXAMPLE 17

5 Simplify each expression.

a	$y \times y \times y \times y$	**b**	$a \times a \times a$
c	$2 \times c \times c$	**d**	$t \times t \times t \times t \times t$
e	$q \times q \times q \times 7$	**f**	$k \times k \times k \times l \times l$
g	$m \times m \times m \times n \times n$	**h**	$-5 \times p \times p \times p \times p \times k \times k \times k$
i	$-3 \times y \times y \times y$	**j**	$d \times d \times (-1) \times e \times e \times f$
k	$q \times q \times r \times r \times r \times r \times 4$	**l**	$3p + 2 \times 4p$
m	$10m \times 4n - 5mn$	**n**	$7a^2 - 2a \times 3a$
o	$6r \times 2y - y \times 10r$	**p**	$5p - 4p \times 8$

6 Write an algebraic expression involving a product that simplifies to: R C

a $18de$ **b** $-20a^2y$ **c** $5pr^2$

Foundation Standard Complex

5.09 Dividing terms

Video
Dividing terms

Dividing terms

When **dividing algebraic terms**, divide the numbers first, then divide the variables.
Write all variables in alphabetical order.

Worksheets
Simplifying algebraic expressions

Why aren't they the same?

Example 18

Simplify each quotient.

a $20p \div 5$ **b** $\frac{24y}{8y}$ **c** $\frac{-2m}{4mn}$ **d** $12a^2b \div 3a$

Solution

a $20p \div 5 = \left(\frac{20}{5}\right)p$ Group numbers and variables separately.
$= 4p$

b $\frac{24y}{8y} = \left(\frac{24}{8}\right)\frac{\cancel{y}}{\cancel{y}}$
$= 3$

c $\frac{-2m}{4mn} = \left(\frac{-2}{4}\right)\frac{\cancel{m}}{\cancel{m}n}$
$= \left(\frac{-1}{2}\right)\frac{1}{n}$
$= -\frac{1}{2n}$

d $12a^2b \div 3a = \left(\frac{12}{3}\right)\frac{\cancel{a}ab}{\cancel{a}}$
$= 4ab$

EXERCISE 5.09 ANSWERS ON P. 576

Dividing terms U F R C

EXAMPLE 18

1 Simplify each quotient.

a $6a \div 3$ **b** $6m \div 3m$ **c** $\frac{15ap}{5p}$

2 Simplify $\frac{55m^2n}{5m}$. Select the correct answer **A**, **B**, **C** or **D**.

A $11mn$ **B** $11m^2n$ **C** $50mn$ **D** $11n$

3 Simplify each quotient.

a $12m \div 3$ **b** $36e \div 4$ **c** $21p \div (-3)$
d $\frac{81x}{9}$ **e** $\frac{60w}{15}$ **f** $\frac{8m}{m}$
g $-25k \div 5k$ **h** $30y \div 40y$ **i** $28pq \div 14q$
j $\frac{63a}{-9a}$ **k** $\frac{4pq}{20q}$ **l** $\frac{-56mn}{8m^2}$

Foundation Standard Complex

9780170477963

4 Simplify each quotient.

a $14m^2 \div 7$ **b** $6e^2 \div (-2e)$ **c** $-5p^2 \div p$

d $\frac{14x^2}{2x}$ **e** $\frac{6wm}{-3m}$ **f** $\frac{8m^2n}{2mn}$

g $32k^2y \div (-4k)$ **h** $9jk^2 \div 12jk^2$ **i** $\frac{15d^2e^2}{3de}$

j $\frac{3a^2b^2c}{3ab^2c}$ **k** $\frac{-24acb}{3bc}$ **l** $\frac{-10m^2n}{5mn}$

5 Simplify $12x \div (-3) \times 2x$. Select **A**, **B**, **C** or **D**.

A $-2x$ **B** -2 **C** $-8x$ **D** $-8x^2$

6 Simplify each expression.

a $-18pq \div 6pq$ **b** $48p \div (-8) \div 2p$ **c** $y \times 3y \times (-5y)$

d $9d \times 4e \div 6d$ **e** $8x \times 9x \div 12$ **f** $48m^2 \div 12m \times 2n$

g $\frac{15c}{5c}$ **h** $\frac{a \times 3ab}{ab}$ **i** $\frac{2ab}{10bc}$

j $\frac{3d \times 4e}{6e}$ **k** $\frac{15mn^2p}{3m^2n^2}$ **l** $\frac{-4x^2y \times 3y}{2xy \times (-6xy)}$

m $6x + 4x \div 2$ **n** $14r \div r + 9r$ **o** $24p - \frac{18p^2}{3p}$

7 Write an algebraic expression involving a quotient that simplifies to: R C

a $7m$ **b** $-3ab$ **c** $5pr^2$

8 Simplify each expression.

a $6w^2 - \frac{10w^5}{5w^3}$ **b** $x^4 - \frac{x^7}{x^3}$ **c** $a \times 2 + 10a \div 5$

d $16m \div 4 - 3 \times 5m$ **e** $12a^3 \div 6a + 15a \times 3a$ **f** $7 \times h + 3 \times w - 4 \times h$

g $14m \div \frac{28m^2n}{4mn}$ **h** $32y \div 16 - 4 \div y + 3 \times y$

5.09

Foundation Standard Complex

5.10 Equations

An **equation** is a statement involving a variable (such as x), numbers and an equals (=) sign, for example, $4x + 5 = 57$. When we find the correct value that makes the equation true, we **solve** the equation. The value is called the **solution** to the equation.

There are different ways of solving equations. The simplest method is the 'guess, check and improve' method.

Worksheet
Guess-and-check

The 'guess, check and improve' method

The 'guess, check and improve' method involves:

- guessing a number for the solution
- checking the guess by substituting the number into the equation
- improving on the guess by testing better numbers until the correct solution is found

Example 19

Use 'guess, check and improve' to solve the equation $4x + 5 = 57$ for x.

Solution

Guess	Check	Comment
$x = 3$	$4 \times 3 + 5 = 17$	Smaller than 57. Try a bigger number.
$x = 10$	$4 \times 10 + 5 = 45$	Still smaller than 57. Try a bigger number.
$x = 20$	$4 \times 20 + 5 = 85$	Bigger than 57. Try a number between 10 and 20.
$x = 15$	$4 \times 15 + 5 = 65$	Still bigger than 57. Try a number between 10 and 15.
$x = 13$	$4 \times 13 + 5 = 57$	Correct.

The solution is $x = 13$.

EXERCISE 5.10 ANSWERS ON P. 577

Equations U F

EXAMPLE 19

1 Solve each equation.

a $x + 3 = 6$ **b** $x - 3 = 6$ **c** $a + 12 = 17$

d $m - 5 = 15$ **e** $b + 1 = 11$ **f** $5c = 20$

g $4k = 24$ **h** $d - 3 = 10$ **i** $10 + m = 25$

j $14 - x = 9$ **k** $\frac{y}{3} = 5$ **l** $\frac{n}{5} = 5$

Foundation Standard Complex

9780170477963

2 What is the solution to $2x - 6 = 24$? Select the correct answer **A**, **B**, **C** or **D**.

A $x = 12$ **B** $x = 15$ **C** $x = 14$ **D** $x = 9$

3 Solve each equation.

a $2x + 4 = 14$ **b** $3p - 5 = 16$ **c** $4k + 6 = 26$

d $5x - 9 = 11$ **e** $7x + 6 = 27$ **f** $8x - 30 = 34$

g $\frac{m}{4} - 3 = 4$ **h** $\frac{a}{2} + 12 = 18$ **i** $10 - 3d = 4$

j $12 - 5n = 7$ **k** $\frac{r-9}{2} = 3$ **l** $\frac{k+1}{3} = 4$

m $2p + 3 = -1$ **n** $\frac{r}{5} + 4 = 6$ **o** $\frac{2y}{3} = 10$

p $2 \times (x + 1) = 6$

4 Solve the equation $6\left(\frac{x}{2} - 1\right) = 12$. Select **A**, **B**, **C** or **D**.

A $x = \frac{13}{3}$ **B** $x = 6$ **C** $x = 3$ **D** $x = 2$

One-step equations

5.11

The 'guess, check and improve' method is simple but slow. There are 2 **algebraic** methods for solving equations that are faster and more efficient:

balancing: performing the same operation on both sides of the equation



Shutterstock.com/defpicture

backtracking: undoing each operation by performing the inverse (opposite) operation



Shutterstock.com/rnl

Puzzle
Equations dominoes

Worksheet
One-step equations

Foundation Standard Complex

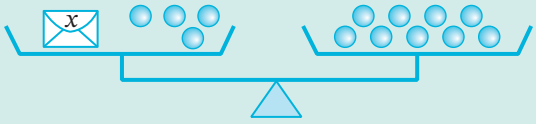
Example 20

Solve the equation $x + 4 = 9$.

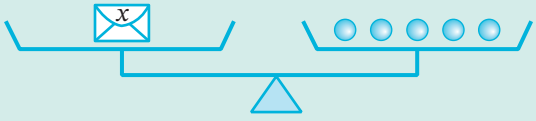
Solution

Method 1: The balancing method

Let x represent an unknown number in an envelope, and ● represent 1.
We can represent the equation $x + 4 = 9$ on balance scales like this:



The 2 sides of the balance are equal. To find the value of x, we can remove 4 balls from both sides.



The solution is $x = 5$.

This equation can also be solved algebraically without diagrams.

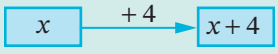
$x + 4 = 9$

$x + 4 - 4 = 9 - 4$ Subtracting 4 from both sides

$\boldsymbol{x = 5}$ Check: $5 + 4 = 9$

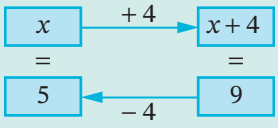
Method 2: The backtracking method

First use a flowchart to show how we get from x to $x + 4$.



In this equation, $x + 4 = 9$. To backtrack (get back) to x, we need to undo the operation 'add 4,' which is 'subtract 4'.

Using a reverse flowchart:



Undoing what has been done to x.

The solution is $\boldsymbol{x = 5}$.

This equation can also be solved algebraically without flowcharts.

$x + 4 = 9$

$x = 9 - 4$ Undoing '+ 4' by subtracting 4

$\boldsymbol{x = 5}$ Check: $5 + 4 = 9$

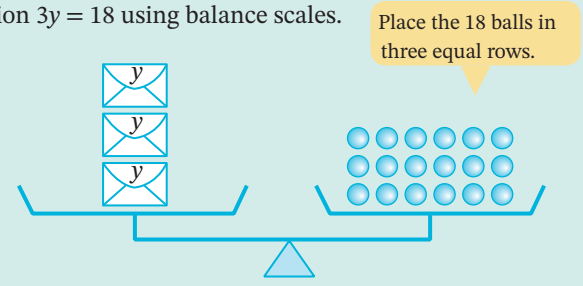
Example 21

Solve $3y = 18$.

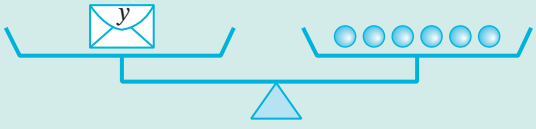
Solution

Method 1: The balancing method

Represent the equation $3y = 18$ using balance scales.



To find the value of y, divide both sides by 3.



The solution is $\boldsymbol{y = 6}$.

Solving this algebraically, we have:

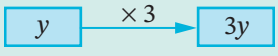
$3y = 18$

$\frac{3y}{3} = \frac{18}{3}$ Dividing both sides by 3

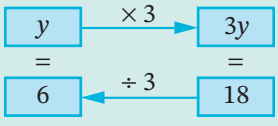
$\boldsymbol{y = 6}$ Check: $3 \times 6 = 18$

Method 2: The backtracking method

First use a flowchart to show how we get from y to $3y$.



In this equation, $3y = 18$. To backtrack to y, we need to undo the operation '$\times$ 3', which is '$\div$ 3', using a reverse flowchart:



The solution is $\boldsymbol{y = 6}$.

Solving this algebraically, we have:

$3y = 18$

$y = \frac{18}{3}$ Undoing '$\times$ 3' in $3y$ by dividing by 3

$\boldsymbol{y = 6}$ Check: $3 \times 6 = 18$

Video
One-step equations

5.11

Solving an equation

When solving an equation, aim to have the variable (such as x) on one side of the equation and a number on the other side, in the form:

$$x = ____$$

Check your solution by substituting it back into the equation.

We call these equations 'one-step' equations because they require one step or one inverse operation to solve.

Example 22

Solve $u - 5 = 7$.

Solution

Method 1: The balancing method

$u - 5 = 7$

$u - 5 + 5 = 7 + 5$ — Add 5 to both sides

$\boldsymbol{u = 12}$ — Check: $12 - 5 = 7$

Method 2: The backtracking method

$u - 5 = 7$

$u = 7 + 5$ — Undoing '-5' by adding 5

$\boldsymbol{u = 12}$ — Check: $12 - 5 = 7$

Example 23

Solve $\frac{k}{7} = 2$.

Solution

Method 1: The balancing method

$\frac{k}{7} = 2$

$\frac{k}{7} \times 7 = 2 \times 7$ — Multiply both sides by 7

$\boldsymbol{k = 14}$ — Check: $\frac{14}{7} = 2$

Method 2: The backtracking method

$\frac{k}{7} = 2$

$k = 2 \times 7$ — Undoing '$\div 7$' by multiplying by 7

$\boldsymbol{k = 14}$ — Check: $\frac{14}{7} = 2$

Inverse operations

Operation	Inverse operation
+	−
−	+
×	÷
÷	×

EXERCISE 5.11 ANSWERS ON P. 577

One-step equations U F R C

EXAMPLE 20

1 Solve each equation, showing the working. R C

a $w + 8 = 15$ **b** $x + 2 = 20$ **c** $m + 3 = 19$

d $p + 12 = 30$ **e** $x + 13 = 22$ **f** $k + 11 = 20$

g $5 + m = 2$ **h** $6 + y = 16$ **i** $15 + d = 6$

EXAMPLE 21

2 Solve each equation, showing the working. R C

a $3m = 18$ **b** $5n = 20$ **c** $4k = 44$

d $2c = 32$ **e** $9x = 81$ **f** $3x = -27$

g $7d = 42$ **h** $6h = -30$ **i** $10a = 70$

EXAMPLE 22

3 Solve each equation, showing the working. R C

a $p - 3 = 8$ **b** $m - 11 = 2$ **c** $x - 5 = 12$

d $y - 20 = 40$ **e** $k - 21 = 3$ **f** $n - 3 = 18$

g $7 = d - 11$ **h** $y - 13 = -4$ **i** $-5 = m - 9$

EXAMPLE 23

4 Solve each equation, showing the working. R C

a $\frac{m}{3} = 4$ **b** $\frac{d}{2} = 5$ **c** $\frac{x}{5} = 8$ **d** $\frac{k}{4} = 12$

e $\frac{x}{10} = 6$ **f** $\frac{a}{2} = -4$ **g** $15 = \frac{k}{2}$ **h** $-11 = \frac{n}{5}$

5 Solve $\frac{p}{3} = -5$. Select the correct answer **A**, **B**, **C** or **D**.

A $p = 15$ **B** $p = 8$ **C** $p = -8$ **D** $p = -15$

6 Solve each equation, showing the working. R C

a $m - 3 = 16$ **b** $3 + p = 11$ **c** $24 = p + 5$

d $n + 7 = -1$ **e** $3x = 30$ **f** $n - 14 = -5$

g $3x = 21$ **h** $4y = -18$ **i** $\frac{h}{7} = 1$

j $-2x = -6$ **k** $6 + k = -4$ **l** $\frac{n}{4} = 0$

7 Solve $3 - m = -2$, showing all working. Select **A**, **B**, **C** or **D**. R C

A $m = -5$ **B** $m = 5$ **C** $m = -1$ **D** $m = 1$

Foundation Standard Complex

5.12 Two-step equations

Videos
Two-step equations 1
Two-step equations 2
Solving equations

Puzzle
Solving equations

Worksheets
Equations writing activity
Equations match
Backtracking 1
Backtracking 2

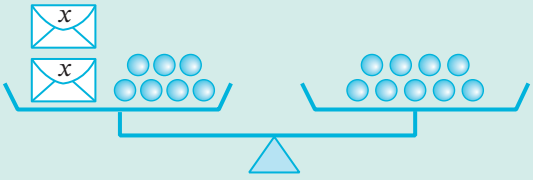
Now we will look at 'two-step' equations that require 2 steps or 2 inverse operations to solve.

Example 24

Solve the equation $2x + 7 = 9$.

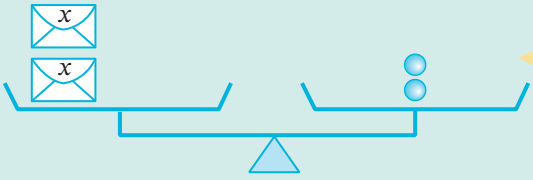
Solution

Method 1: The balancing method



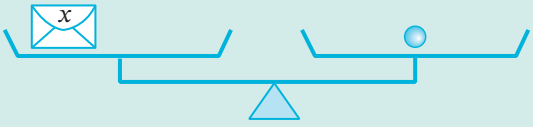
We can represent $2x + 7 = 9$ using balance scales.

First subtract 7 balls from both sides:



Place the remaining balls into 2 equal rows.

Then divide both sides by 2.



The solution is $\boldsymbol{x = 1}$.

Using algebra we have:

$2x + 7 = 9$

$2x + 7 - 7 = 9 - 7$ Step 1: Subtracting 7 from both sides

$2x = 2$

$\frac{2x}{2} = \frac{2}{2}$ Step 2: Dividing both sides by 2

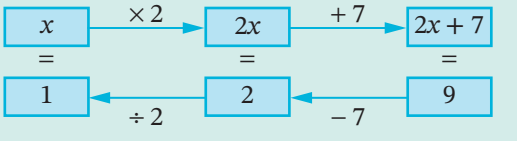
$\boldsymbol{x = 1}$

Check: $2 \times 1 + 7 = 9$

Method 2: The backtracking method

Use a flowchart to go from x to $2x + 7$.

$2x + 7 = 9$. To backtrack, we need to undo the operations '$\times$ 2'and '$+$ 7' in reverse order, so '$-$ 7' and '$\div$ 2' using a reverse flowchart.



9780170477963

The solution to $2x + 7 = 9$ is $x = 1$.

Using algebra, we have:

$2x + 7 = 9$

$2x = 9 - 7$ Step 1: Undoing '+ 7' by subtracting 7

$2x = 2$

$x = \frac{2}{2}$ Step 2: Undoing '× 2' in $2x$ by dividing by 2

$\boldsymbol{x = 1}$ Check: $2 \times 1 + 7 = 9$

Note that inverse operations are performed in **reverse order**. For example, to undo putting on our socks and shoes, we take off our shoes first, then our socks.

Example 25

Solve $\frac{M}{3} + 2 = 5$.

Solution

Method 1: The balancing method

$\frac{M}{3} + 2 = 5$

$\frac{M}{3} + 2 - 2 = 5 - 2$ Step 1: Subtracting 2 from both sides

$\frac{M}{3} = 3$

$\frac{M}{3} \times 3 = 3 \times 3$ Check: $\frac{9}{3} + 2 = 3 + 2 = 5$

$\boldsymbol{M = 9}$ Step 2: Multiplying both sides by 3

Method 2: The backtracking method

$\frac{M}{3} + 2 = 5$ Step 1: Undo '+ 2' by subtracting 2

$\frac{M}{3} = 5 - 2$

$\frac{M}{3} = 3$

$M = 3 \times 3$ Step 2: Undo '÷ 3' by multiplying by 3

$\boldsymbol{M = 9}$ Check: $\frac{9}{3} + 2 = 3 + 2 = 5$

EXERCISE 5.12 ANSWERS ON P. 577

Two-step equations U F R C

1 Solve each equation showing all steps. Remember to check your answers. R C

EXAMPLE 24

a	$2x + 1 = 11$	**b**	$3x + 8 = 17$	**c**	$4x + 5 = 13$
d	$4x + 9 = 25$	**e**	$5x + 2 = 47$	**f**	$2x + 10 = -6$
g	$-3x + 10 = 4$	**h**	$-4x + 9 = 5$	**i**	$-5x + 8 = -12$
j	$-7x + 4 = -17$	**k**	$-2x + 7 = 9$	**l**	$-4x + 14 = 22$

Foundation Standard Complex

2 The following is Liam's incorrect solution for $7x + 5 = 13$. R

$7x + 5 = 13$

$7x = 13 - 5$	**A**
$7x = 8$	**B**
$x = \frac{8}{7}$	**C**
$x = 1\frac{1}{8}$	**D**

In which line was the error made? Select **A**, **B**, **C** or **D**.

3 Solve each equation, showing all steps. R C

a	$2x - 5 = 9$	**b**	$3x - 1 = 23$	**c**	$5x - 3 = 12$
d	$2x - 4 = 18$	**e**	$6x - 4 = 8$	**f**	$5x - 19 = -9$
g	$4x - 7 = 5$	**h**	$7x - 3 = 32$	**i**	$2x - 4 = -14$
j	$5x - 2 = 43$	**k**	$3x - 4 = -10$	**l**	$8x - 12 = -20$

4 What is the solution for $\frac{n}{4} - 8 = 7$? Select **A**, **B**, **C** or **D**. R

A $n = 4$ **B** $n = 20$ **C** $n = 36$ **D** $n = 60$

EXAMPLE 25

5 Solve each equation. R C

a	$\frac{x}{2}+5=9$	**b**	$\frac{m}{3}+6=9$	**c**	$\frac{k}{5}+2=6$
d	$\frac{a}{4}+4=8$	**e**	$\frac{n}{2}+6=14$	**f**	$\frac{h}{7}+2=-3$
g	$\frac{m}{2}-8=-5$	**h**	$\frac{k}{5}-7=-4$	**i**	$\frac{x}{3}-6=-10$
j	$\frac{a}{3}-1=2$	**k**	$\frac{h}{5}-4=-7$	**l**	$\frac{x}{2}-3=4$

6 Solve $\frac{k+1}{3} = 5$, showing all working. Select **A**, **B**, **C** or **D**. R C

A $k = 19$ **B** $k = 18$ **C** $k = 17$ **D** $k = 14$

7 Solve each equation. R C

a	$\frac{x}{2}+x=1$	**b**	$\frac{x}{2}+\frac{x}{3}=1$	**c**	$\frac{3}{m}+\frac{4}{m}=14$

Quiz
Mental skills 5B

Mental skills 5B: Maths without calculators ANSWERS ON P. 577

Multiplying by 9, 11 and 12

To multiply a number by 9, multiply by 10 and then subtract the number.
This is because 10 times a number minus the same number equals 9 times the number.

1 Study each example.

a $14 \times 9 = 14 \times 10 - 14 = 140 - 14 = 126$

b $25 \times 9 = 25 \times 10 - 25 = 250 - 25 = 225$

c $18 \times 9 = 18 \times 10 - 18 = 180 - 18 = 162$

Foundation Standard Complex

9780170477963

2 Now simplify each product.

a 12×9 **b** 27×9 **c** 46×9 **d** 19×9

e 34×9 **f** 63×9 **g** 21×9 **h** 15×9

To multiply a number by 11, multiply by 10 and then add the number.
This is because 10 times a number plus the same number equals 11 times the number.

3 Study each example.

a $26 \times 11 = 26 \times 10 + 26 = 260 + 26 = 286$

b $13 \times 11 = 13 \times 10 + 13 = 130 + 13 = 143$

c $35 \times 11 = 35 \times 10 + 35 = 350 + 35 = 385$

4 Now simplify each product.

a 17×11 **b** 22×11 **c** 38×11 **d** 40×11

e 25×11 **f** 19×11 **g** 54×11 **h** 31×11

To multiply a number by 12, multiply by 10, then add double the number.
This is because 10 times a number plus double the same number equals 12 times the number.

5 Study each example.

a $22 \times 12 = 22 \times 10 + 22 \times 2 = 220 + 44 = 264$

b $16 \times 12 = 16 \times 10 + 16 \times 2 = 160 + 32 = 192$

c $70 \times 12 = 70 \times 10 + 70 \times 2 = 700 + 140 = 840$

6 Now simplify each product.

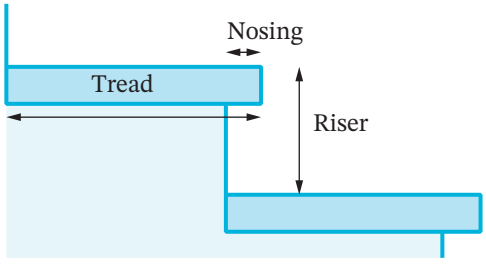
a 44×12 **b** 15×12 **c** 29×12 **d** 31×12

e 52×12 **f** 18×12 **g** 26×12 **h** 37×12

Did you know?

Taking safe steps

Staircases must be designed to be safe to use; they must provide enough stepping space and not be too steep. The diagram below shows 2 steps. The **tread** is the length of a step, and the **riser** is the height between the tops of steps. The **nosing** is how much the tread of one step 'overhangs' horizontally with the step below it.



The Australian Building Code recommends that risers must be between 13 cm and 22.5 cm and the tread must be between 21.5 cm and 35.5 cm.

The French architect François Blondel developed an equation to design a comfortable and efficient staircase:

$$2 \times \text{Riser} + \text{Tread} = \text{a number from 63 to 65}$$

For example, suppose we need to design a staircase that is 2.6 m high.

1 Calculate the number of steps that will be needed

Assuming an ideal riser of 18 cm, the number of steps will be:

$260 \text{ cm} \div 18 \text{ cm} = 14.44 \ldots \approx 15$ steps

2 Calculate the height of each riser

$\text{Riser} = 260 \text{ cm} \div 15 = 17.33 \ldots \approx 17$ cm

3 Calculate the tread

Apply the Blondel formula:

$2 \times \text{Riser} + \text{Tread} = 64$

$2 \times 17 + \text{Tread} = 64$

Solving the equation:

$34 + \text{Tread} = 64$

$\text{Tread} = 30$ cm

The staircase will have 15 steps, each with a 30 cm tread and 17 cm riser.

Now calculate the measurements for a staircase that is 3.2 m high.

5.13 Equation problems

Puzzle
Writing and solving equations

Video
Equation problems 1

Many mathematical problems are stated in words. We can solve these problems by translating the problems into algebraic symbols and equations.

Example 26

When a number is doubled and then 5 is added, the answer is 69. Find the number.

Solution

Let the number be x.

Translating the problem into an equation:

$x \times 2 + 5 = 69$

$2x + 5 = 69$

Solve the equation:

$2x + 5 - 5 = 69 - 5$ Subtracting 5 from both sides

$2x = 64$

$\frac{2x}{2} = \frac{64}{2}$ Dividing both sides by 2

$x = 32$ Check: $2 \times 32 + 5 = 69$

The number is 32.

Example 27

Tom is 8 years older than Susi. If the sum of their ages is 22, find their ages.

Solution

Let Susi's age be n.

So Tom's age is $n + 8$.	Tom is 8 years older
$n + n + 8 = 22$	Susi's age + Tom's age = 22
$2n + 8 = 22$	Simplifying $n + n$ to $2n$
$2n + 8 - 8 = 22 - 8$	Subtracting 8 from both sides
$2n = 14$	
$\frac{2n}{2} = \frac{14}{2}$	Dividing both sides by 2
$n = 7$	Check: $2 \times 7 + 8 = 22$
So Susi is 7 and Tom is $7 + 8 = 15$	Check: $7 + 15 = 22$

Videos
Equation problems 1
Equation problems 2

5.13

Solving word problems using equations

- Choose a variable
- Translate the problem into an equation
- Solve the equation
- Write a sentence that answers the problem

EXERCISE 5.13 ANSWERS ON P. 577

Equation problems U F PS R

EXAMPLE 26

1 Solve each problem by writing an equation and then solving it. You may use diagrams to help you think about the information. PS R

a 5 tickets for a film cost \$55. How much does each ticket cost?
(Let t represent the price of a ticket.)

b 10 oranges cost \$4.80. How much does each orange cost?
(Let x represent the cost of one orange.)

c A number is doubled and the result is 110. What is the number?
(Let n represent the unknown number.)

d A number has 4 subtracted from it and the result is 6. Find the number.
(Let y represent the unknown number.)

2 For each word problem, select the correct equation **A**, **B**, **C** or **D**. Then solve the equation to solve the problem. PS R

a Jarrad has collected 1794 beetles. This is 6 times as many beetles as Lisa has in her collection. How many beetles does Lisa have?

A $1794 - N = 6$ **B** $6N = 1794$

C $N = 6 \times 1794$ **D** $N + 6 = 1794$

Foundation Standard Complex

b Kurt mixed 590 mL of white paint with some blue paint. He mixed 1.73 L of paint altogether. How much blue paint did he use?

A $N + 590 = 1730$ **B** $N - 590 = 1730$

C $\frac{N}{590} = 1730$ **D** $590N = 1730$

c 14 packets of chocolate biscuits are packed in a box.
The supermarket sold 546 packets of biscuits. How many boxes were sold?

A $14N = 546$ **B** $546 - N = 14$

C $N + 14 = 546$ **D** $\frac{N}{14} = 546$

d When a certain number is subtracted from 100, the result is 47. What is the number?

A $N - 100 = 47$ **B** $100 + 47 = N$

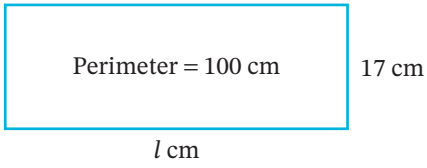
C $N - 47 = 100$ **D** $100 - N = 47$

EXAMPLE 27

3 Translate each of the following problems into an equation, then solve the equation to solve the problem. PS R

a Mr Abdul says, 'If you multiply my age by 4 and add 12, the answer is 240.' How old is Mr Abdul? (Let a stand for his age.)

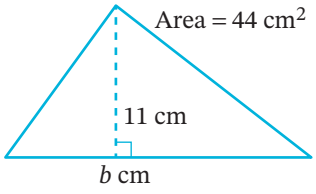
b The perimeter of a rectangle is 100 cm and its width is 17 cm. What is the length? (Let l cm represent the length of the rectangle.)



c I think of a number. If I multiply it by 6 and subtract 13, the answer is 95. What is the number? (Let y represent the number.)

d The sum of 3 consecutive numbers is 48. Find the smallest number. (Let n be the smallest number.)

e The area of a triangle is calculated by multiplying the base by the height and dividing by 2. If a triangle has an area of 44 cm^2 and its height is 11 cm, what is the length of its base? (Let b cm represent the length of the base.)



f The Student Council is holding a school disco to raise money. Each ticket bought by students raises $12, but the cost of running the disco is $600. How many tickets must be sold to make a profit of $2040? (Let n stand for the number of tickets sold.)

g Grace earns $400 per week plus $\frac{1}{5}$ of the value of her gym equipment sales for that week. If she is paid $1000 for one week, what is the value of her sales for that week? (Let x stand for the value of her sales.)

Foundation Standard Complex

9780170477963

4 A repairer of washing machines charges using the formula: PS R

$$C = 75h + 140$$

where C is the charge in dollars and h is the number of hours the job takes.

Find the number of hours worked if the charge is \$515.

5 The weekly profit, in dollars, made by a beach hire stall is given by the formula $P = 5u - 900$, where u represents the number of beach umbrellas hired. Find the number of umbrellas hired if the profit is \$1055. PS R

6 The volume of a rectangular prism is given by the formula $V = lwh$. What is the value of l when $V = 340$, $w = 4$ and $h = 5$? Select **A**, **B**, **C** or **D**.

A 17 **B** 68 **C** 85 **D** 425

7 Julian, Marcus and Aiden compared how many merit awards they received in class. Marcus was awarded 7 more than Julian, but Aiden had as many as Marcus and Julian put together. Altogether they collected 54 merit awards. Let a be the number of awards Julian had. How many awards did Aiden receive? Select **A**, **B**, **C** or **D**. PS R

A 10 **B** 17 **C** 20 **D** 27

8 There are 50 boys and girls in a club. At the end of the year, each boy contributed \$6 towards the cost of the end of year party. The girls contributed \$5 each. In total, they raised \$270. Let g be the number of girls. How many boys and girls are in the club? Select **A**, **B**, **C** or **D**. PS R

A 20 boys, 30 girls **B** 34 boys, 16 girls

C 30 boys, 20 girls **D** 25 boys, 25 girls

Investigation

School concert

Year 7 wants to hold a school concert to help subsidise the cost of their camp. They want to raise \$1500 for this.

The concert will cost \$100 for each night it is on. The hall will hold 250 people. If a band is hired for the concert, this would cost an extra \$550 per night.

a If Year 7 decide to stage one concert only without a band, calculate how much each ticket should be to cover the costs and raise the required amount. Use an equation to help you.

b If they decide to have the band, write another equation to find the new ticket price.

c If the SRC decides to have 3 concerts with the band, calculate what the ticket price should be. Consider how many people you think would attend each night. Write another equation and solve it to justify your answer.

Power plus ANSWERS ON P. 577

1 Simplify each expression.

a $2a + b + 3a - 4b$ **b** $m - 3n - 5m - 7n$

c $3k - 2j - 5 - 4k + 6j + 9$ **d** $a^2 + 2b^2 + 3b^2 - a^2$

e $4x^2 + 6x - x^2 - 12x$ **f** $xy + yz + xz + 4xy + 3xz + yz$

2 Evaluate each expression if $p = 4$, $q = -6$, $r = 3$ and $t = -1$.

a $p + q$ **b** pq **c** $r^2 + t^2$

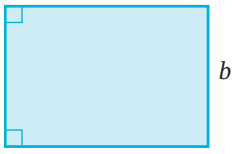
d $q^2 - 6p + rt$ **e** $\frac{pq}{rt}$ **f** $6 - 2r + p$

g $2p + 3q + 4r - 5t$ **h** $\frac{p+4}{2}$

3 For each shape, write an algebraic expression for:

i its perimeter **ii** its area

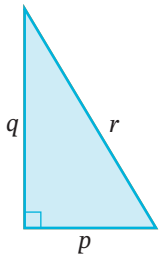
a



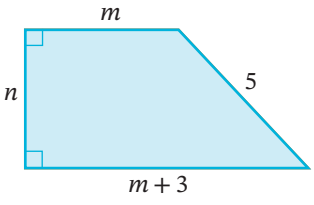
b



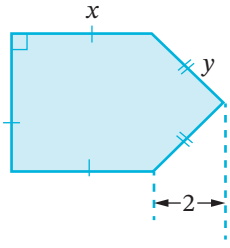
c



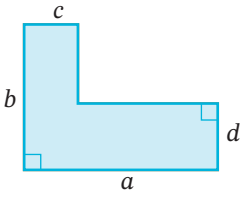
d



e



f



4 If x is any number, then simplify each expression.

a $x + 0$ **b** $x \times x$ **c** $0 \div x$ **d** $x \div x$

e $x - (-x)$ **f** $0 \times x$ **g** $0 - x$ **h** $(-x)^2$

i $x \times \frac{1}{x}$ **j** $x + \frac{1}{x}$

5 Solve each equation.

a $6 - 3x = 9$ **b** $9 - 8k = 33$ **c** $7 - 2n = 11$

d $2(r - 4) = 10$ **e** $3(d + 6) = 18$ **f** $4(z + 1) = -6$

 9780170477963

CHAPTER 5 REVIEW

Language of maths

abbreviation	algebraic term	associative law	backtracking
balancing	check	commutative law	consecutive
distributive law	equation	evaluate	expanded form
expression	formula	guess, check and improve	inverse
like terms	one-step	pronumeral	solution
solve	substitute	two-step	variable

1 What is another name for **pronumeral**?

2 What is the difference between an **expression** and an **equation**?

3 Which law involves multiplying by a number by splitting that number to make the calculation easier?

4 What are **consecutive numbers**?

5 What are **like terms**?

6 What name is given to the value that makes an equation true?

Quiz
Language of maths 5

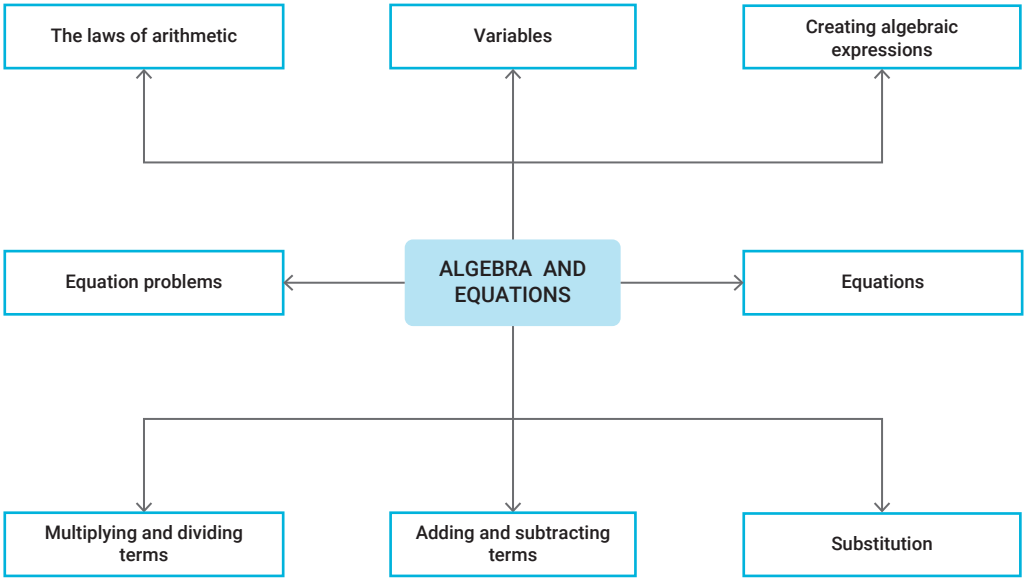
Puzzle
Algebra find-a-word

Topic summary

- Which parts of this topic did you find easy? Was there anything you already knew?
- Are there parts of this chapter that you still do not understand? Talk to your teacher.
- Give 2 examples of jobs where algebra or equations might be needed.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Algebra and equations



Quiz
Test yourself 5

TEST YOURSELF 5

ANSWERS ON P. 578

5.01 **1** Use the commutative and associative laws to evaluate each expression.

a $7 + 18 + 3$ **b** $91 + 18 + 9$ **c** $2 \times 17 \times 5$

d $25 \times 31 \times 4$ **e** 13×30 **f** 7×400

5.02 **2** Use the distributive law to evaluate each expression.

a 22×9 **b** 17×8 **c** 27×11 **d** 13×12

5.03 **3** Examine this number pattern, then write the general rule using a variable.

$3 - (-3) = 2 \times 3$

$5 - (-5) = 2 \times 5$

$11 - (-11) = 2 \times 11$

$2 - (-2) = 2 \times 2$

$8 - (-8) = 2 \times 8$

5.03 **4** Simplify each expression.

a $x + x + x + x$ **b** $u \div 5$ **c** $2y - y$

d $5 \times a \times 2$ **e** $8 \times c \times b \times 3 \times a$ **f** $2 \times m \times w \times 10 \times w$

5.03 **5** Write each of these in expanded form.

a $4mnk$ **b** $\frac{6a}{b}$ **c** $9c^2$

5.04 **6** Write an algebraic expression for each statement. Use N to represent any number.

a 3 times the number **b** the difference between the number and 5

c the next consecutive number **d** one-third of the number

5.04 **7** Write an expression for:

a the sum of M and 3 **b** 5 more than B **c** $2H$ decreased by k

5.05 **8** Find the value of these expressions if $a = 2$, $b = 5$ and $c = 6$.

a $a + b$ **b** $3a - c$ **c** $a + 2b - c$ **d** abc

5.05 **9** If $C = 30h + 19$, find C when

a $h = 3$ **b** $h = -1$ **c** $h = 5$ **d** $h = 2.5$

5.07 **10** Simplify each expression by collecting like terms.

a $3m + 4m$ **b** $6uv + 4 + 12uv$ **c** $5h^2 + 6h + 8h^2 + 11h$

d $12m - 4m$ **e** $7y - 4y - y$ **f** $13h^3 - 6h - 8h - 11h^3$

5.08 **11** Simplify each expression.

a $3m \times 4m$ **b** $9v \times 4 \times 5v$ **c** $3pq \times 5pq$

d $a \times a \times a \times a$ **e** $-7d \times (-3d)$ **f** $lmn \times lmn \times lmn$

Foundation Standard Complex

9780170477963

12 Simplify each expression. 5.09

a $15m \div 3$ **b** $9v \div v$ **c** $\frac{49y}{7}$

d $\frac{12x^2y}{4xy}$ **e** $32de^2 \div 10de$ **f** $\frac{-3x^2y \times 2x}{3x \times (-x)}$

13 Solve each equation using the 'guess, check and improve' method. 5.10

a $4m - 9 = 7$ **b** $\frac{k+2}{7} = 2$ **c** $\frac{d}{3} - 4 = 2$

14 Solve each equation algebraically, showing all steps. 5.11

a $w + 17 = 52$ **b** $e - 12 = 0$ **c** $5p = 15$ **d** $\frac{x}{2} = 6$

15 Solve each equation algebraically, showing all steps. 5.12

a $3t + 4 = 10$ **b** $2n + 6 = 12$ **c** $4y - 7 = 5$

d $\frac{m}{4} - 8 = 9$ **e** $\frac{x}{3} + 5 = 8$ **f** $\frac{m-4}{2} = 3$

16 **a** When a certain number is multiplied by 17, the product is 1003. Which equation can be used to solve this problem? (Let N represent the number.) 5.13
Select the correct answer **A**, **B**, **C** or **D**.

A $N + 17 = 1003$ **B** $1003 - N = 17$

C $17N = 1003$ **D** $1003 + N = 17$

b Solve the equation to solve the problem.

17 Lara organises a charity raffle at her school. The only cost is $180 for prizes. 5.13

a If she sets the price of raffle tickets at $5, write an equation to find how many tickets she must sell to just cover costs, and solve the equation. (Let N represent the number of tickets.)

b To make a profit of $2100, how many tickets must she sell?



Alamy Stock Photos/The Photolibrary Wales

Foundation Standard Complex